solution methods of heterogeneous agent models are solved this way by constructing a representative agent with special weights, where the weights do not necessarily correspond to wealth or income.[13] The weights on some agents may be zero. Agents with small proportions of total wealth in heterogeneous agent models can have very large effects in determining prices.

In this context, the representative agent should not be interpreted as the average agent, but as the *marginal* agent. Prices are set through how the marginal agent responds to small changes in factor shocks, which is different from how the average agent responds. For the asset owner weighing optimal equity allocations, the context of the previous sections—whether bad times for me correspond to bad times for the average agent or whether my risk aversion increases when the risk aversion of the average agent increases—are identical except that we ask the questions in the context of the agents determining prices at the margin. For example, can I absorb losses during bad times more than the asset managers forced to liquidate at fire-sale prices? Am I exposed to the same margin calls as hedge funds?

In heterogeneous agent models, a new factor arises that determines risk premiums. Asset prices now depend on the cross-sectional distribution of agent characteristics. This can be the cross-sectional distribution (across agents) of wealth, beliefs, labor income shocks, or other variables in which investors differ.[14] In Constantinides and Duffie (1996), agents exhibit heterogeneity by how much they earn. They also cannot completely hedge the risk of job loss (as in the real world). Constantinides and Duffie show that if income inequality increases during recessions (more formally, the cross-sectional variance of idiosyncratic labor income shocks increases), as it does in data, the equity premium will be high. Intuitively, in recessions, the probability of job losses increases and equities drop in value. That is, equities are poor hedges for insuring against the risk of losing your job and won't pay off when your boss has fired you and you need to eat. These considerations make equities quite unattractive and mean that equities must exhibit a high return in equilibrium to induce investors to hold them.

The income shocks that workers face in the Constantinides and Duffie world are an example of *market incompleteness*. This refers to risks that cannot be hedged or removed in aggregate and thus, in effect, serve to increase the total amount of risk that all agents (or the implied representative agent) face. A similar effect can be engineered by constraints, such as those on borrowing.[15] During bad times, some investors may face binding funding constraints that force them to behave

[13] These are called Negishi (1960) weights.

[14] Models with heterogeneity in risk aversion, which give rise to the factor of a cross-sectional wealth distribution, were developed by Dumas (1989) and Wang (1996). For models with heterogeneity in beliefs, see David (2008).

[15] See, among many others, Constantinides and Duffie (1996) and Constantinides, Donaldson, and Mehra (2002).